

Name: _____

_____ / 25

Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

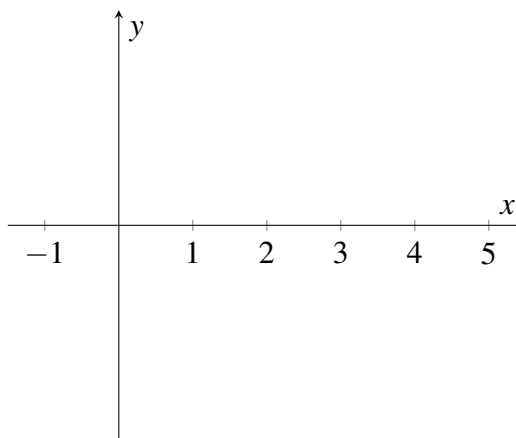
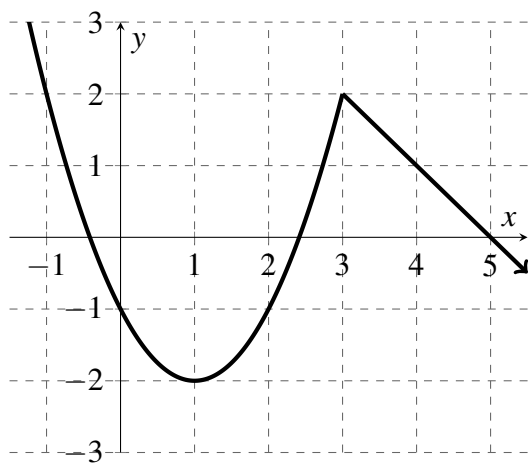
1. [8 points] Consider the function $f(x) = \frac{2}{3-x}$.

Find $f'(1)$ using the limit definition: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

(Finding the derivative using any other derivative method will result in 0 points for this problem.)

$$f'(1) =$$

2. [5 points] Using the graph of $g(x)$ below, sketch a graph of **the derivative** $g'(x)$ on the axis to the right.



3. [12 points] Find the derivative of each function. You may use any method you like, and you do not need to simplify your answer.

a. $f(x) = 4x^7 - 5x^6 + \frac{x}{6} - 7$

$$f'(x) =$$

b. $g(x) = \frac{30 + x - x^4}{x^2}$

$$g'(x) =$$

c. $h(x) = \cos(x) (x^4 + x^3 + x + 1)$

$$h'(x) =$$

d. $p(x) = \sqrt{x} (x^2 + \cos(x)) + 2^4$

$$p'(x) =$$